

Views

ARTIFICIAL INTELLIGENCE

Tech companies should stop pretending AI won't destroy jobs

No matter what anyone tells you, we're not ready for the massive societal upheavals on the way.

I took an Uber to an artificial-intelligence conference at MIT one recent morning, and the driver asked me how long it would take for autonomous vehicles to take away his job. I told him it would happen in about 15 to 20 years. He breathed a sigh of relief. "Well, I'll be retired by then," he said.

Good thing we weren't in China. If a driver there had asked, I would have had to tell him he'd lose his job in about 10 years—maybe 15 if he was lucky.

That might sound surprising, given that the US is, and has been, in the lead in AI research. But China is catching up—if it hasn't already—and that rivalry, with one nation playing off the other, guarantees that AI is coming.

China will have at least a 50/50 chance of winning the race, and there are several reasons for that.

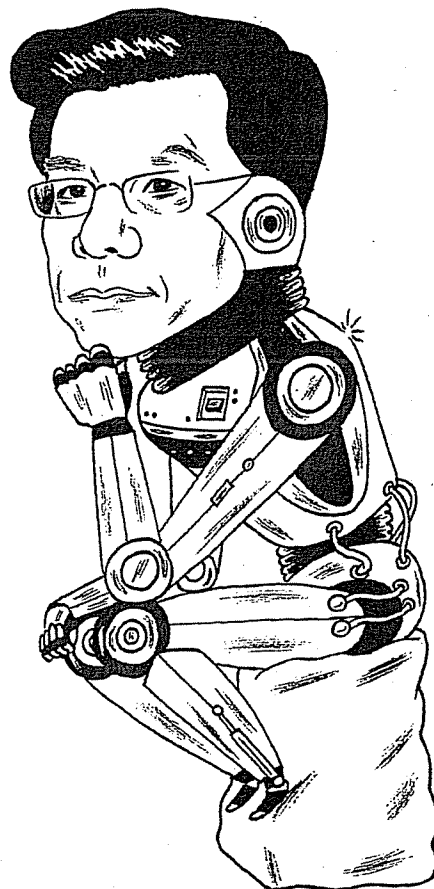
First, China has a huge army of young people coming into AI. Over the past decade, the number of AI publications by Chinese authors has doubled. Young AI engineers from Face++, a Chinese face-recognition startup, recently won first place in three computer-vision challenges—ahead of teams from Google, Microsoft, Facebook, and Carnegie Mellon University.

Second, China has more data than the US—way more. Data is what makes AI go. A very good scientist with a ton of data will beat a super scientist with

a modest amount of data. China has the most mobile phones and internet users in the world—triple the number in the United States. But the gap is even bigger than that because of the way people in China use their devices. People there carry no cash. They pay all their utility bills with their phones. They can do all their shopping on their phones. You get off work and open an app to order food. By the time you reach home, the food is right there, hot off the electric motor-bike. In China, shared bicycles generate 30 terabytes of sensor data in their 50 million paid rides per day—that's roughly 300 times the data being generated in the US.

Third, Chinese AI companies have passed the copycat phase. Fifteen years ago almost every decent startup in China was simply copying the functionality, look, and feel of products offered in the US. But all that copying taught eager Chinese entrepreneurs how to become good product managers, and now they're on to the next stage: exceeding their overseas counterparts. Even today, Weibo is better than Twitter. WeChat delivers a way better experience than Facebook Messenger.

And fourth, government policies are accelerating AI in China. The Chinese government's stated plan is to catch up with the US on AI technology and applications by 2020 and to become a global AI innovation hub by 2030. In a speech in October, President Xi Jinping encouraged further integration of the internet, big data, and artificial intelligence with the real-world economy. And in case you're wondering, these things tend not to be all talk in China—as demonstrated with its past policies promoting high-speed rail and the mass entrepreneurship and innovation movement. In comparison, things get bogged down in the US. Consider the way President Barack Obama's loan guarantee to solar-panel



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maker Solyndra was hammered as crony capitalism. Truckers are now appealing to President Donald Trump and Congress to stop testing of autonomous trucks.

The rise of China as an AI superpower isn't a big deal just for China. The competition between the US and China has sparked intense advances in AI that will be impossible to stop anywhere. The change will be massive, and not all of it

Then there are the symbiotic optimists, who think that AI combined with humans should be better than either one alone. This will be true for certain professions—doctors, lawyers—but most jobs won't fall in that category. Instead they are routine, single-domain jobs where AI excels over the human by a large margin.

Others think we'll be saved by a universal basic income. "Take the extra

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good. Inequality will widen. As my Uber driver in Cambridge has already intuited, AI will displace a large number of jobs, which will cause social discontent. Consider the progress of Google DeepMind's AlphaGo software, which beat the best human players of the board game Go in early 2016. It was subsequently bested by AlphaGo Zero, introduced in 2017, which learned by playing games against itself and within 40 days was superior to all the earlier versions. Now imagine those improvements transferring to areas like customer service, telemarketing, assembly lines, reception desks, truck driving, and other routine blue-collar and white-collar work. It will soon be obvious that half of our job tasks can be done better at almost no cost by AI and robots. This will be the fastest transition humankind has experienced, and we're not ready for it.

Not everyone agrees with my view. Some people argue that it will take longer than we think before jobs disappear, since many jobs will be only partially replaced, and companies will try to redeploy those displaced internally. But even if true, that won't stop the inevitable. Others remind us that every technology revolution has created new jobs as it displaced old ones. But it's dangerous to assume this will be the case again.

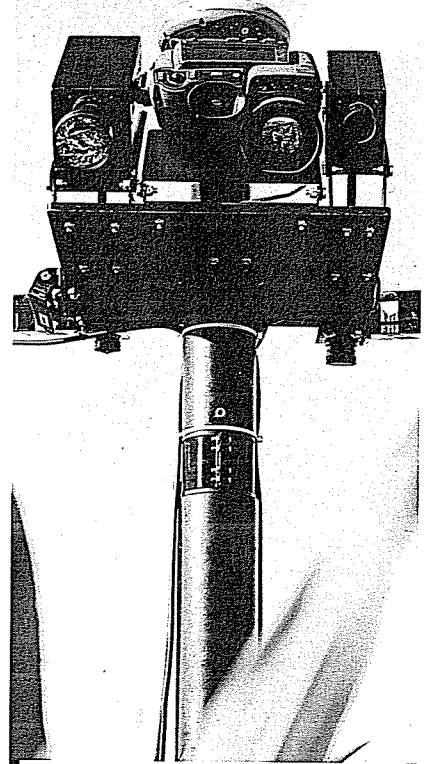
money made by AI and distribute it to the people who lost their jobs," they say. "This additional income will help people find their new path, and replace other types of social welfare." But UBI doesn't address people's loss of dignity or meet their need to feel useful. It's just a convenient way for a beneficiary of the AI revolution to sit back and do nothing.

And finally, there are those who deny that AI has any downside at all—which is the position taken by many of the largest AI companies. It's unfortunate that AI experts aren't trying to solve the problem. What's worse, and unbelievably selfish, is that they actually refuse to acknowledge the problem exists in the first place.

These changes are coming, and we need to tell the truth and the whole truth. We need to find the jobs that AI can't do and train people to do them. We need to reinvent education. These will be the best of times and the worst of times. If we act rationally and quickly, we can bask in what's best rather than wallow in what's worst.

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